

● INTELLIGENT VALUE CHAIN ECOSYSTEM · EST 2026

Ultra cold chain operations, *end-to-end* *controlled.*

One platform. Six modules. Every shipment.
Every degree. Always in view — from dispatch
sign-off to ULT transfer at the receiving dock.

- Cold Chain Ultra Cold Chain Vaccines Reagents
- Quality Control Samples Biologics Compounds



TRUCKINGCUBE · SIX PILLARS, ONE ECOSY

 Open Dashboard View Control Tower**-80°C****FLOOR**ULT operating
envelope**<15****MIN**Alert
acknowledgement
SLA**68****CONTROLS**Across dispatch
· transit ·
delivery**5 yr****RETENTION**Audit-grade
record archive**CAPABILITY STACK**

Built for ultra cold chain.

Six pillars of infrastructure and readiness — every shipment is engineered before it moves. From -80 °C truck-boxes to a control tower that never sleeps, every link is

owned, audited, and instrumented.



01

Reefer Trucks & Portable Coldboxes

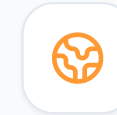
- ✓ Dual-zone reefers + IATA passive coldboxes, -80 °C to +25 °C
- ✓ Battery backup, calibrated probes, on-board IoT loggers
- ✓ Pre-cool SOP & door-to-door temperature continuity



02

Transit Support & Control Tower

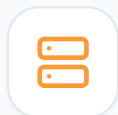
- ✓ 24×7 live IoT temp + GPS dashboard
- ✓ Alert SLA <15 min · escalation matrix wired
- ✓ Driver hotline · safe-haven dry-ice top-ups



03

Air · Road · Rail Network

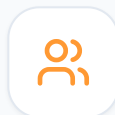
- ✓ Air freight DGR-2 partners across BOM·DEL·BLR·HYD
- ✓ Dual-zone reefer fleet on national corridors
- ✓ Reefer rail nodes for long-haul efficiency



04

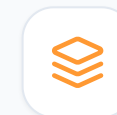
IT Platform & Stack

- ✓ TMS · QMS · WMS unified on a single platform



05

Compliance & QA Officer



06

Storage Warehouse

- ✓ Multi-zone: ULT -80, frozen, 2-8 °C, ambient

- ✓ IoT loggers · GPS · alerting · audit trail
- ✓ API-ready for shipper, customer, regulator

- ✓ Dedicated QA Manager + dispatch QA officer
- ✓ NCR & deviation workflow inside QMS
- ✓ CMVR · IATA DGR · Schedule M sign-off owner

- ✓ Mapped, qualified & 24×7 monitored
- ✓ Schedule M + WHO PQS audit-ready

SINGLE SOURCE OF TRUTH

One platform. Six modules.

TruckingCube unifies SOP execution, transit telemetry, delivery quality gates, and alert escalation under a single audit-grade platform.



SOP Library

01

Versioned phase checklists with proof requirements



Process Engine

02

Dispatch · Transit · Delivery state machine



Transit Tracker

03

Live IoT telemetry, GPS, dry-ice top-ups

**Delivery Gate**

04

Pre-acceptance checks · ULT
transfer in 10 min

**Alert & Escalation**

05

Drift · excursion · tamper
triggers, ladder timer

**Audit-Grade Trail**

06

Every state change retained
for 5 years

THREE PHASES · 68 CONTROLS

From release gate to ULT transfer, every step measured.

Each shipment follows three sealed phases with explicit gates, owners, and proof requirements. Excursions raise an NCR within four hours.

PHASE 01

Dispatch

22 controls · dual sign-off



- ✓ Container pre-cool to -78°C verified
- ✓ Logger armed, baseline within $\pm 2^{\circ}\text{C}$
- ✓ Dry-ice charge per 1.5× sublimation rule
- ✓ DG paperwork · UN1845 · IATA DGR-2 packed
- ✓ QA Manager + Warehouse Supervisor sign-off
- ✓ Sealed container photographed with timestamp

PHASE 02

Transit

22 controls · escalation triggers



- ✓ Continuous logger uplink every 2 min
- ✓ Drift / excursion engine – 4 trigger types
- ✓ Cross-dock dwell ≤ 5 min, CoC seal verified
- ✓ Dry-ice top-ups at scheduled safe havens
- ✓ 15-min ack ladder · driver $\rightarrow$ CT $\rightarrow$ LM
- ✓ Tamper photo evidence captured per handoff

PHASE 03

Delivery

24 controls · ULT in 10 min



- ✓ Receiver notified ≥ 60 min before arrival
- ✓ Logger pre-acceptance gate before container open
- ✓ Open container photo with timestamp
- ✓ ULT transfer completed within 10 min
- ✓ Receiving QA sign-off · NCR if excursion
- ✓ All records sealed into 5-year audit archive



Excursions trigger immediate quarantine.

Outside -58 °C / -82 °C raises a HIGH severity alert and an NCR within four hours. The shipment is held until QA disposition.

[View Escalation Matrix →](#)